Lab 7 Fuzz Testing Report

Input Validation Bugs Found with Hypothesis

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Result

Hypothesis fuzzing exposed defects in all three processor functions. The initial run had 11 failing tests and 8 passing tests. After input validation and edge-case fixes, all 19 tests passed.

Functions and fixes

normalize_text now validates string input and collapses every Unicode whitespace sequence. **parse_integer_list** accepts blank input as an empty list, validates input types, and reports malformed tokens clearly. **summarize_numbers** rejects non-integer values and returns a defined summary for an empty list.

Test results

Run	Result	What it showed
Buggy baseline	11 failed, 8 passed	Whitespace, types, empty inputs
Fixed processor	19 passed	All fuzz properties and validation tests pass

Reproduce

```
python -m pip install -r requirements.txt
python processor.py
python -m pytest test_processor.py -q
```

The destination repository did not include a Lab 7 starter processor.py. This submission therefore includes a self-contained three-function example and a matching pre-fix baseline so the fuzzing steps can be reproduced.

The evidence panels are rendered from the saved command output; complete raw output is included as text files in the evidence folder.

Step 2 | Buggy application run

```
BUGGY APPLICATION RUN

Normalized text: Ada    Lovelace
Parsed integers: [12, -3, 9]
Summary: {'count': 3, 'min': -3, 'max': 12, 'total': 18, 'average': 6.0}
```

The pre-fix program runs on typical inputs but leaves the tab between words unchanged.

Step 4 | Initial fuzz failures

```
INITIAL FUZZ TEST FAILURES - KEY OUTPUT

FFFFFFFFF.....F...F                                                    [100%]
E   AssertionError: assert 'Ada\tLovelace' == 'Ada Lovelace'
E   AttributeError: 'NoneType' object has no attribute 'strip'
E   ValueError: invalid literal for int() with base 10: ''
E   ZeroDivisionError: division by zero
E   Failed: DID NOT RAISE TypeError
11 failed, 8 passed, 1 warning in 8.34s
```

The initial test run found 11 failures; the panel shows key minimized cases and the test summary.

Step 5 | Fixed application run

```
FIXED APPLICATION RUN

Normalized text: Ada Lovelace
Parsed integers: [12, -3, 9]
Summary: {'count': 3, 'min': -3, 'max': 12, 'total': 18, 'average': 6.0}
```

The fixed program normalizes the tab and keeps the expected integer summary.

Step 6 | Final fuzz test output

```
FINAL HYPOTHESIS FUZZ TESTS

..................                                                [100%]
19 passed, 1 warning in 3.01s
```

The complete updated suite passed: 19 tests.